JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

# FPGA Design Using VHDL

Module 5A: Simulators Revisited

# This Module, You Will Learn About…

1. Simulator Operations
   - Differentiate between **event** and **transaction**
   - Create a testbench containing File I/O
2. Misc. Topics
   - Shift Registers
   - Debouncing Logic

# VHDL Simulation Time

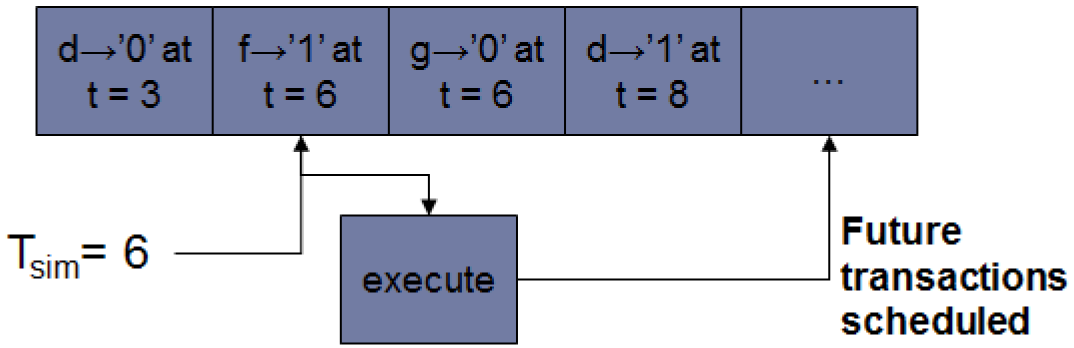
- There is one simulation time for the entire system under test
- Simulation is indexed by an integer Tc
- Units are in seconds
  - Default resolution specified when simulator invoked
    - 1 ns for functional simulation
    - 100 ps for timing simulation
- Simulation time begins at 0 s

# VHDL Simulation Terminology

- A transaction is an update of the current value of an object
  - Value may not change
- An object is active if it experiences a transaction
- An event is a transaction that results in a change of value
- Event $\Rightarrow$ Transaction
- A process is executed when there is a event on any of the objects in the sensitivity list
  - Remember from Module 2: A process is executed when there is a “change” on any of the objects in the sensitivity list
  - Replace “change” with “event”

# Event Driven Simulation

- VHDL simulator is a **transaction** driven simulator
- Simulator queues **transactions** scheduled for the future
  - Processed in chronological order
  - Each **transaction** can schedule future transactions, which are added to the queue



# VHDL Simulation Terminology Example

- Example of transaction versus event
- Students, tell me if `c` and `d` experience an event or transaction

```
process(a, b)
begin
    c <= a and b;
end process;
```

```
process(c)
begin
    d <= not c;
end process;
```

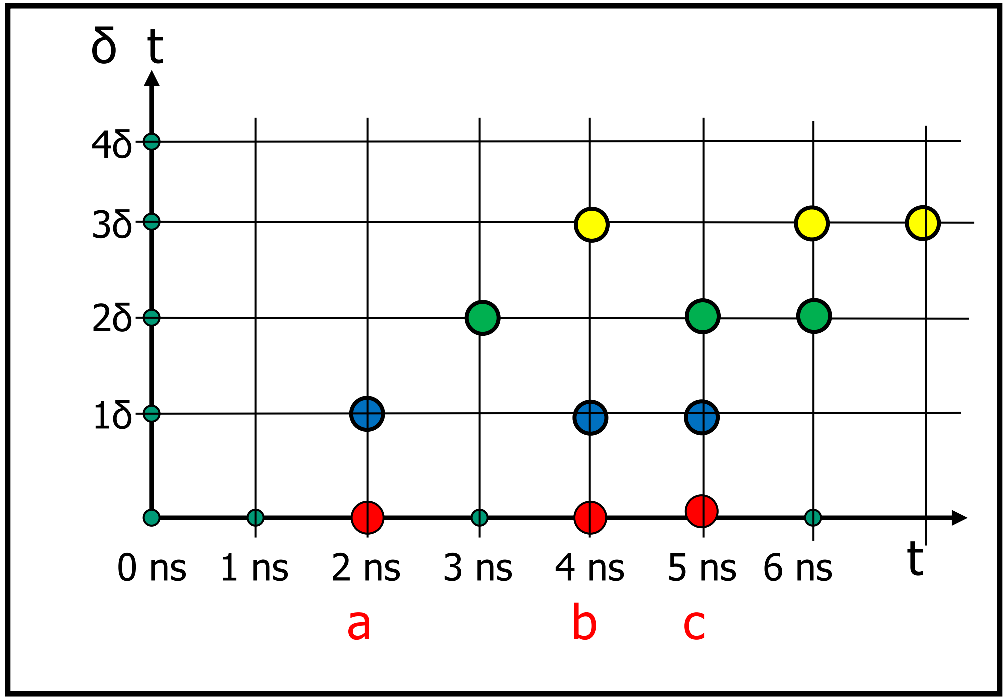
| Sim Time | a | b | c | d |
|---|---|---|---|---|
| 0 | u | u | u (event) | u (event) |
| 100 ns | 0 | 0 | 0 (event) | 1 (event) |
| 200 ns | 0 | 1 | 0 (transaction) | 1 (transaction) |
| 300 ns | 1 | 0 | 0 (transaction) | 1 (transaction) |
| 400 ns | 1 | 1 | 1 (event) | 0 (event) |

# Delta Time

- Infinitely small (δ) advance in time
- The current simulation time TC does not advance as δ advances
    - TC + δ = TC
    - TC + 100 δ = TC
- Infinite number of delta time steps can occur during current simulation time TC
- Simulator places statements in order
- Signals/ports in process updated at end of process
- Variables updated immediately

# More on Delta Time

```
                a                b                c
0    x <='1' after 2 ns, '0' after 4ns, '1' after 5 ns;
1δ   y <= x after 0 ns;
2δ   z <= x after 1 ns;
3δ   w <= z after 1 ns;
```




Scheduled Events
(or transactions)

- x
- y
- z
- w